

Spatial ML - class 14 - spatial association rules

Total points 100/100 ?

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What distinguishes spatial association rules from standard association rules?

*10/10

- ☐ They require time-series data
- ☒ They include information about spatial context or neighbourhoods
- ☐ They only use distance matrices
- ☐ They only work with categorical variables



Why are continuous variables transformed into labeled intervals? ^{*} 10/10

- ☐ To increase randomness
- ☐ To remove spatial information
- ☐ To reduce dataset size
- ☒ To allow usage in association rule mining algorithms

Which of the following is NOT a way to include spatial information? ^{*} 10/10

- ☐ Measuring distance to important locations
- ☐ Clustering nearby observations
- ☐ Using neighbourhood characteristics
- ☒ Ignoring location completely



What is the main purpose of thresholding variables? *

10/10

- ☐ Normalize distributions
- ☒ Convert continuous data into interpretable categories
- ☐ Increase model speed only
- ☐ Remove noise completely

In R, what does the following code do? *

10/10

```
trans1 <- as(data_basket, "transactions")
```

- ☐ Filters variables
- ☐ Calculates distances
- ☒ Converts dataset into transaction format for association rules
- ☐ Converts data into a spatial object



What does the apriori() function do? *

10/10

- ☐ Performs clustering
- ☒ Generates association rules from transactional data
- ☐ Visualizes spatial data
- ☐ Computes regression

What is the purpose of the appearance argument in apriori()? *

10/10

- ☐ To filter missing data
- ☒ To restrict which items appear on LHS or RHS of rules
- ☐ To set confidence thresholds
- ☐ To visualize rules



When is a rule considered redundant? ^{*} 10/10

- ☐ When it has low support
- ☐ When it has too many variables
- ☒ When a more general rule exists with equal or higher confidence
- ☐ When it includes spatial features

What is the purpose of testing rule significance? ^{*} 10/10

- ☐ To increase the number of rules
- ☒ To ensure rules are statistically meaningful (e.g., via Fisher's test)
- ☐ To remove all spatial variables
- ☐ To normalize data



Which of the following best describes “lift” in association rules? ^{*} 10/10

- ☐ Number of items in a rule
- ☒ Strength of dependence compared to random co-occurrence
- ☐ Frequency of occurrence
- ☐ Distance between observations

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